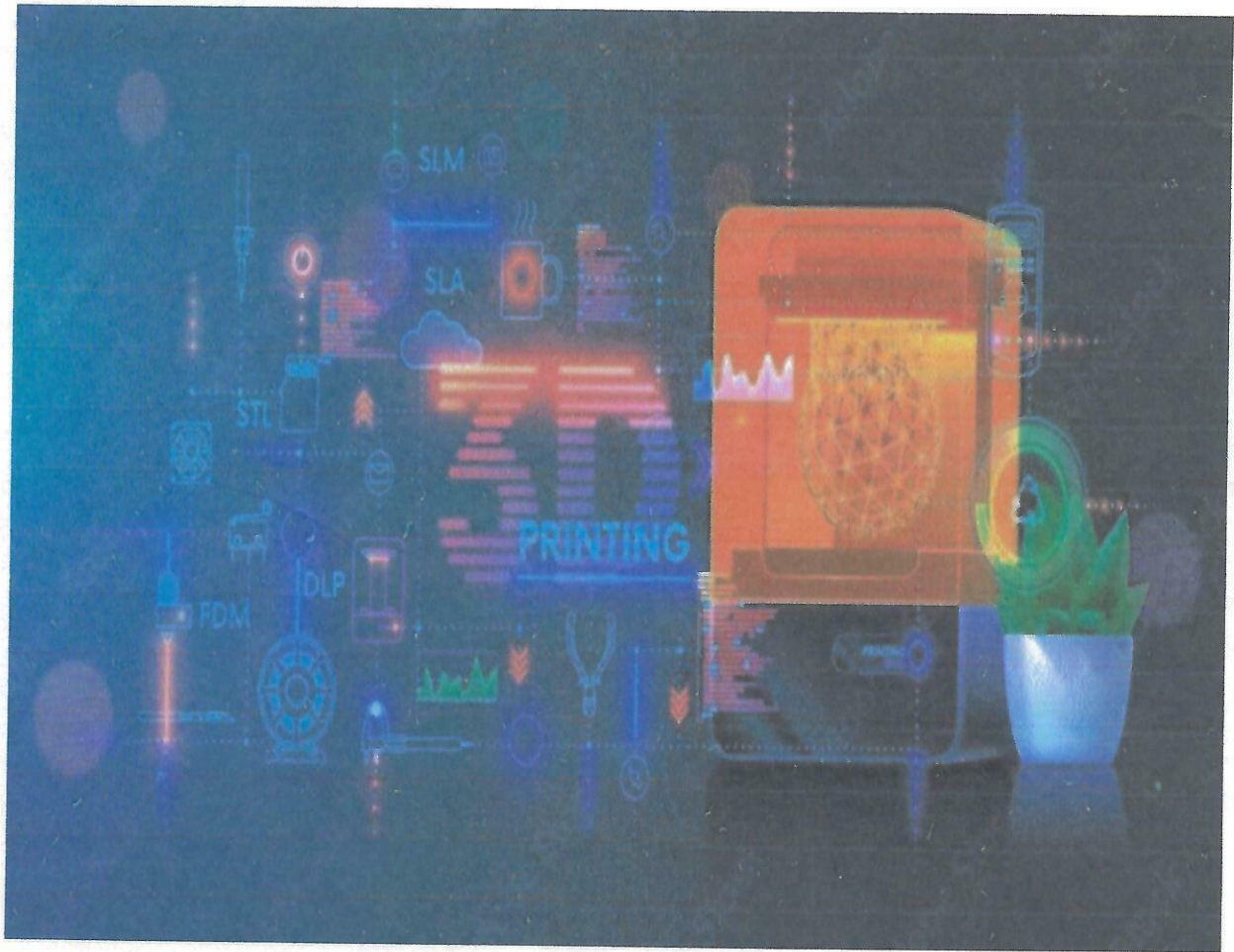


3D PRINTING REPORT



Certificate of Approval

It is hereby approved that the project report entitled **Report of 3D Printing** submitted by:

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As part of their certification-based industrial training in **3D Printing Technology**, conducted under the supervision of **WTL & MeitY, Cooch Behar Government Engineering College**, Cooch Behar, West Bengal.

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13/08/25

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REPORT FOR 3D PRINTING

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Group -A

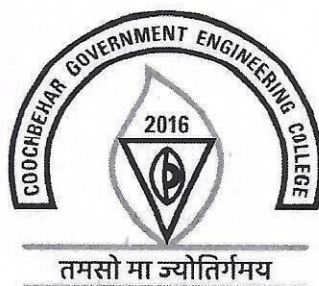
Under the guidance of :

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Cooch Behar Government Engineering College

2025

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected faculty members, **Dr. PALASH DAS** (HOD of ECE Dept., CGEC), **Mrs. SUSMITA BANIK BARIK** (Project Eng., WTL & MeitY Project), **Mr. ARIJIT BHUNIA** (Project Co-ordinator, WTL & MeitY Project) and **Mr. BHASKAR MONDAL** (Faculty, WTL & MeitY Project) for their valuable guidance and constant support throughout these projects.

We find it extremely necessary to express our gratitude to the WTL & MeitY Project, to our institute and the Govt. of India for giving us this amazing opportunity to get a hands-on training on 3D Printing at our own college. Also we feel very grateful to them, that they have arranged such an engaging environment for our 3D Printing lab in our own college with Air Conditioners, Smart-Board, Top-notch computers and different kinds of 3D printers.

We also thank to all our group members for their collaborative work and endless support. From designing to manufacturing, they have show an enthusiastic team spirit, a deterministic attitude and sleepless hardwork.

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OBJECTIVE

The objective of this report is to provide an in-depth understanding of the 3D printing technology, its various techniques, and the materials used in the process. The report aims to explore the different 3D printing methods, such as FDM, SLA, SLS, and DLP, and highlight their respective advantages, limitations, and ideal use cases. Additionally, the report will discuss common file formats used in 3D printing, such as STL and OBJ, and the associated printing parameters, including layer resolution, speed, and material type.

Furthermore, the report will showcase a 3D-printed object developed by the team, detailing its design, the materials used, and the printing process followed to bring it to completion. The report will also evaluate the specific choices of printing parameters, such as temperature, print speed, and layer height, and analyze their impact on the final quality of the printed object. Ultimately, the goal of this report is to provide a comprehensive overview of 3D printing technology, demonstrating both the technical and practical applications of this innovative manufacturing process.

INTRODUCTION TO 3D PRINTING

3D printing, also known as **additive manufacturing**, is a process of creating three-dimensional objects from a digital file. Unlike traditional manufacturing methods, which often involve subtracting material (e.g., cutting, drilling), 3D printing builds objects layer by layer from the ground up. This allows for the creation of complex, customized, and intricate designs that would be difficult or impossible to achieve with traditional techniques. Though some of the 3D Printing techniques use subtractive method in some cases.

Additive Process: In 3D printing, material is added in successive layers. These layers are precisely deposited based on a digital design. This is different from subtractive manufacturing methods, where material is cut away from a block.

- **Materials used** : Plastic, Metals, Ceramic, Liquid, Powder etc.
- The object to be printed is created digitally using computer-aided design (CAD) software; Then that design is sliced by a slicer software, where we also set other printing parameters.
- Most often a post-processing is also needed on the printed object for better finishing.

Applications of 3D Printing :

- **Prototyping:** 3D printing allows rapid prototyping, helping designers and engineers create functional prototypes quickly and inexpensively.
- **Healthcare:** Custom prosthetics, dental implants, and surgical tools can be printed to fit an individual's needs.

- **Manufacturing:** 3D printing can create small, intricate parts for machinery or consumer products. It's increasingly being used for low-volume production.

Advantages of 3D Printing :

- ✓ **Customization:** 3D printing allows for easy customization, enabling unique, one-of-a-kind designs.
- ✓ **Speed:** It enables rapid prototyping, which can reduce time to market for products.
- ✓ **Complexity:** Complex geometries that would be difficult or impossible to manufacture using traditional methods can be easily created.

Challenges of 3D Printing :

- **Speed for Large Items:** While 3D printing is fast for small to medium objects, it can take hours or days to print larger items.
- **Material Limitations:** Although there is a broad range of materials, some 3D-printed objects might not be as durable or heat-resistant as those made through traditional methods.
- **Post-Processing:** Many 3D prints require post-processing, such as cleaning or curing, to achieve the final look or finish.

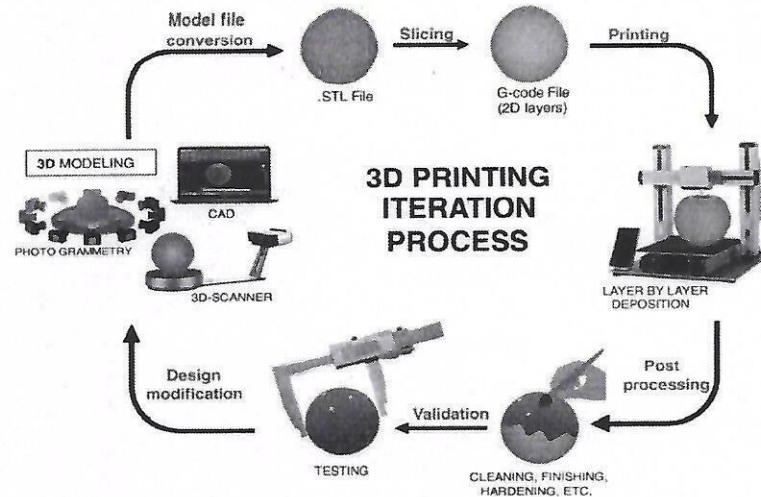
The Future of 3D Printing :

3D printing is expected to continue evolving with advancements in speed, material variety, and application range. Some areas of significant future growth include:

- ❖ **3D Printing in Space:** NASA and other space agencies are researching the use of 3D printing to create tools and structures on Mars and the Moon.
- ❖ **Bioprinting:** Researchers are exploring ways to 3D print tissues, organs, and medical implants, potentially revolutionizing healthcare.
- ❖ **Sustainability:** There's ongoing research into using recyclable materials and creating more eco-friendly printing methods.

JOURNEY FROM DESIGN TO 3D OBJECT

The journey from design to 3D object printing involves several key stages: conceptualizing and creating a digital model using CAD software, preparing the model through slicing for layer-by-layer printing, selecting appropriate materials and printing parameters, and finally, transforming the design into a tangible object through the printing process.



- **CAD Software** : CAD or **Computer-aided design** helps to create the 3D design of the object to be printed from scratch or by reverse-engineering from a scan of any object.
- **Slicing** : The 3D model is then sliced into many layers so that the 3D Printer can understand and manufacture the object.
- **3D Printing** : The sliced model is then printed layer by layer by the 3D Printer.
- **Post-Processing** : Post-Processing includes smoothening, cutting, washing, coloring the manufactured object for better finishing.

3D File Formats

- **Neutral File-format** : They are software-independent and are used to ensure compatibility between different CAD software and 3D printers. Ex : .stl, .STEP, .3mf .
- **Native File-format** : They are the proprietary file types used by specific CAD or 3D modeling software. Native formats are typically not directly used for 3D printing but are crucial during the design phase. Ex : .dwg, .blend, .sldprt etc.

STL File Format :

- From a CAD software, a design is often exported as .stl format for further slicing, as they are easily understood by 3D Printers.
- STL stands for Stereolithography or Standard Tessellation Language or Standard Triangle Language.
- It represents the surface geometry of a 3D object using a collection of linked triangles or polygons.

Key points about Slicing :

- **Converts Model to Layers**: Breaks the 3D model into thin horizontal layers (slices).
- **Generates G-code**: Produces machine instructions (G-code) for the 3D printer.
- **Print Settings**: Includes layer height, infill density, print speed, and support structures.

- **Supports:** Adds temporary structures for overhangs and complex geometries.
- **Material Considerations:** Adjusts settings for different filament types (e.g., PLA, ABS).

3D Printing Settings :

Here's a quick overview of 3D printing settings parameters:

1. **Layer Height:** Controls the thickness of each printed layer (e.g., 0.1mm, 0.2mm). Affects resolution and print speed.
2. **Infill Density:** Percentage of internal structure (e.g., 20%, 50%). Higher density means stronger prints but longer print times.
3. **Print Speed:** Speed at which the printer moves (e.g., 50mm/s).
4. **Temperature (Extruder):** The heat of the nozzle, dependent on filament type (e.g., 200°C for PLA).
5. **Temperature (Bed):** The heat of the print bed (e.g., 60°C for PLA).
6. **Support Structures:** Extra material for printing overhangs .
7. **Shell/Wall Thickness:** Thickness of outer layers , impacts strength and print quality.
8. **Infill Pattern:** The internal geometry pattern of the print (e.g., grid, honeycomb).

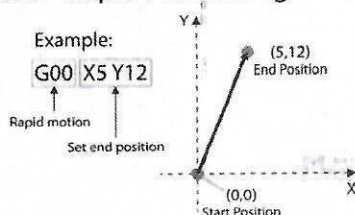
What is G-code ?

- G-code of Geometry-Code is the language that controls 3D printers, telling them how to move, extrude, and print layer by layer.

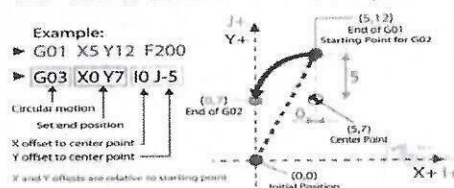
Key points :

- **Definition:** G-code is the language that controls 3D printers, telling them how to move, extrude, and print layer by layer.
- **Generated by Slicer:** G-code is created by slicing software (e.g., Cura, PrusaSlicer) based on your 3D model and print settings.
- **Instructions:** Includes commands for movement (G0/G1), temperature (M104/M109), fan control (M106/M107), and more.

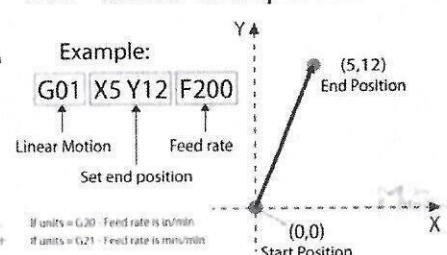
G00 - Rapid Positioning



G03 - Counterclockwise Circular Interpolation



G01 - Linear Interpolation



CLASSIFICATION OF ADDITIVE MANUFACTURING BY FEEDSTOCK STATE

- Solid-based
- Liquid-based
- Powder-based

1. Solid-based processes :

Feed material is a solid (sheet, filament, wire), joined layer-by-layer.

- **Laminated Object Manufacturing (LOM)** – thin sheets laminated, then cut to shape.
- **Ultrasonic Additive Manufacturing (UAM)** – metal sheets ultrasonically welded, then cut.

2. Liquid-based processes :

Material is in liquid or paste form, solidified via curing.

- **Stereolithography (SLA)** – liquid photopolymer cured by laser.
- **Digital Light Processing (DLP)** – similar to SLA but uses a projector.
- **PolyJet / Material Jetting** – jets drops of photopolymer, cured by UV.

3. Powder-based processes :

Uses powder materials, bonded via thermal, binder, or kinetic mechanisms.

3A. Powder Bed Fusion (PBF) :

Thermally fuses powder in a bed:

- **Selective Laser Sintering (SLS)** – sinters plastics/ceramics.
- **Selective Laser Melting / Direct Metal Laser Sintering (SLM/DMLS)** – melts metal powder fully.

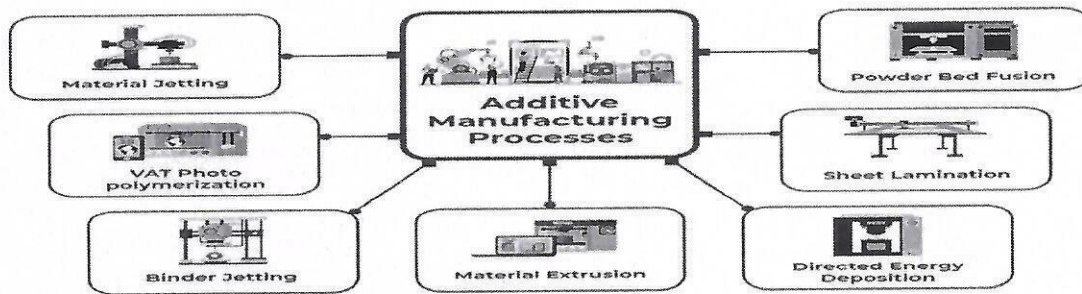
3B. Binder Jetting :

Applies binder to selectively glue powder, followed by sintering or infiltration.

3C. Directed Energy Deposition (DED) / Powder Directed Energy Deposition :

Powder is blown through a nozzle and melted with a laser or other energy source; often used to repair parts.

ADDITIVE MANUFACTURING TECHNIQUES



Material Extrusion :

FDM (Fuse deposition modelling) or its generic term **FFF** (Fuse Filament Fabrication) is a common material extrusion process and is trademarked by the company Stratasys. It works by feeding a continuous thermoplastic filament from a spool into a heated extruder nozzle.

Key Points :

- **Filament types:** Common polymers include **PLA, ABS, PETG, Nylon, TPU**, and even engineering materials like **PC, PEEK, ULTEM**; composites (e.g. carbon-fiber, conductive, dissolvable) are also available.
- **Low cost & accessible:** Equipment and materials (e.g. PLA/ABS filament) are affordable and widely available.

Commonly Used Materials for FDM :

PLA (Polylactic Acid) :

- ✓ **Easy to print:** Low print temperature (~180–220 °C), no heated bed required, minimal warping. Excellent for beginners and fine-detail models.
- ✓ **Eco-friendly:** Made from renewable resources (e.g., corn starch), biodegradable under industrial composting conditions.



Fig : PLA



Fig : ABS

ABS (Acrylonitrile Butadiene Styrene) :

- ✓ **Strength & temperature tolerance:** Good impact resistance, durable, and handles heat up to ~98-100 °C without deformation.
- ✓ **Print challenges:** Requires a heated bed (~80–110 °C) and ideally an enclosure to avoid warping and cracking; emits unpleasant fumes—ventilation needed.

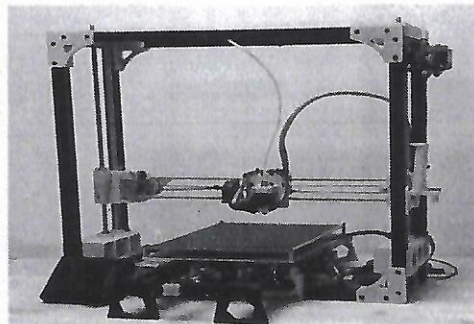
PETG (Polyethylene Terephthalate Glycol-modified) :

- ✓ **Balance of properties:** Combines good tensile strength (~40–50 MPa) with excellent impact resistance and layer adhesion, resulting in durable yet slightly flexible parts.
- ✓ **Moderate heat and chemical resistance:** Performs well up to ~70–80 °C; resistant to water, many solvents, and UV better than ABS.
- ✓ **Printability:** Needs heated bed (~70–90 °C), may string more, but warps less than ABS and often doesn't need enclosure.

Types of FDM 3D Printers

i) Cartesian ii)Delta iii)Polar iv)Hybrid

Cartesian 3D Printers :



How it works:

- **Movement:** The printhead moves along the X, Y, and Z axes (typically, the bed moves in the Z direction, while the extruder moves in the X and Y).
- **Structure:** It has a rectangular frame with three linear rails for movement.

Common Features:

- ✓ **Most Common Type:** This design is the most popular among hobbyist and entry-level machines (e.g., Prusa i3, Creality Ender 3).
- ✓ **Accuracy:** High precision and ease of calibration.
- ✓ **Simplicity:** Simple mechanics, making it cost-effective and reliable for most standard prints.

Examples: i) Prusa i3 MK3 ii) Creality Ender 3

Delta 3D Printers :

How it works:

- **Movement:** The printhead is controlled by three **vertical arms** (Delta arms), moving in a circular pattern, not the traditional linear axes.
- **Structure:** Uses a **triangular frame** with three arms attached to a central extruder.

Common Features:

- ✓ **Fast Printing:** Faster than Cartesian printers in many cases due to its parallel arm design.
- ✓ **Less Rigid:** More susceptible to vibrations and accuracy issues unless precisely calibrated.

Examples:

- Anycubic Kossel
- Robo 3D R1+ Delta

Polar 3D Printers :

How it works:

- **Movement:** Uses a rotating build platform (instead of a standard flat bed) combined with a vertical extruder.
- **Structure:** The object rotates during the print, allowing for printing complex geometries with multiple angles and intricate designs.

Common Features:

- ✓ **Ideal for round objects:** Great for cylindrical or spherical prints, like vases or large, curved parts.
- ✓ **Unique Setup:** Less common and more niche, typically used for specialized industrial purposes.

Examples:

- Zortrax Inventure
- TrioVibes Rotary 3D Printer



Fig : BIGREP STUDIO 3D PRINTER

GRADUATE FROM DESKTOP TO INDUSTRIAL :

The BigRep STUDIO is a high-performance 3D printer with an expansive 1000×500×500mm build volume. Designed for professionals producing professional-grade projects, it features precision-engineered components, dual extrusion, and an advanced filament detection system. Its fast, reliable printing capabilities and user-friendly interface make it ideal for prototyping, production, and design applications.

Key Features of the BigRep STUDIO G2 Industrial Large-Format 3D Printer:

- Large Build Volume: 1000mm×500mm×500mm
- Scale up production while cutting costs without compromising quality
- Fast, reliable printing for efficient production

The BigRep STUDIO Dual Mode comes with:

- (1) BigRep SDX Extruder, featuring (2) SDX Fiber Hotends with 0.6 mm nozzles
- (1) BigRep Polyimide Print Bed Tape
- BigRep BLADE Slicing Software
- BigRep CONNECT for monitoring, analysis, and remote control of your BigRep printer

THE MACHINE THAT SCALES WITH YOU

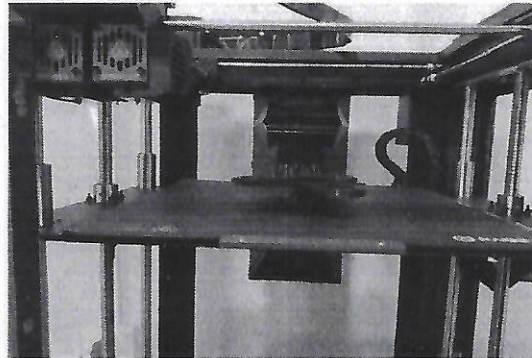
Meet the BigRep STUDIO G2—your gateway to industrial large-format 3D printing. Offering the simplicity of a desktop printer and 10 times the build volume, this compact powerhouse delivers industrial-scale manufacturing capabilities, fitting seamlessly into any workspace.

- Large-format 3D printing for increased output
- Industrial performance and print-quality
- Easy to use as a desktop 3D printer

6. Products Made Using 3D Printing :

In our project, we created several products using both **PLA** and **PETG** filaments.

- Using **PLA**, we printed basic components and prototypes such as:



Hand Spinner (Fidget Spinner) :

We designed and 3D printed a hand spinner using **PLA filament**. The spinner body was modeled in **FreeCAD** with precise dimensions to hold ball bearings. After printing, we inserted bearings into the center and arms to allow smooth rotation.

Wrenches and Generator Handle (Printed with PETG) :

Using **PETG filament**, we 3D printed two **wrenches of different sizes** and a **handle for a generator**. The wrenches were designed to be strong and slightly flexible, making them suitable for light mechanical tasks.

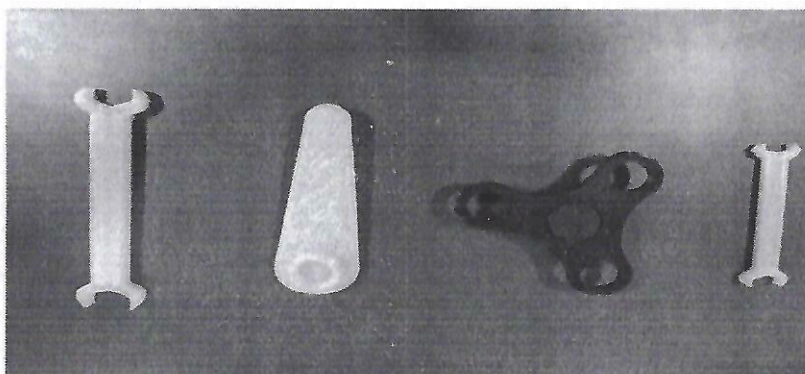


Fig - 3D Printed Wrenches and Generator Handle (Material: PETG)
Fidget Spinner (Material : PLA)

Conclusion

3D printing can offer benefits across the entire creation process from initial concept design to final manufacturing and all steps in between. Different applications have unique needs and understanding those application requirements is critical when choosing a 3D printer. Multiple systems may offer broader use opportunities than a single system, so identifying your unique requirements to apply 3D printing across your entire design-to-manufacture process can shorten time-to-market, improve product performance, streamline and cost-reduce manufacturing, and improve product quality and customer satisfaction will help you define the ideal 3D printing capability for your organization.

We understand how the all the data help in making a decision and in optimizing the process. This experience gave me the importance of documentation and reporting the data as it was essential in tracking performance and making an informed decision.

Last but not least, my internship has provided me with many experience that can enhance my skill such as analytical thinking and technical thinking. All of the skills that I have learned can be benefit to my future career in industry which is I can apply all the things that I learned on it.

In conclusion, industrial training offers a valuable opportunity for students to gain knowledge beyond the academic and technical aspects in a professional working environment. During the training period, I was able to experience the reality of the workplace and see the importance of building positive relationships between employees and trainees. Such interactions contribute to a supportive and collaborative work culture, which is essential for personal and professional growth.